BUSINESS INTELLIGENCE LAB

EXPERIMENT 9

BI Security – Authentication, Authorization and Access Control using Microsoft Power BI

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1. Aim

To study and demonstrate Business Intelligence security by identifying users and roles, analysing authentication methods, designing authorization and access control, implementing Row-Level Security (RLS) in Microsoft Power BI Desktop, and assessing security risks with suitable controls.

2. Introduction

Business Intelligence (BI) systems bring together sensitive information such as sales, profit and customer data. If this information is available to everyone, there is a risk of misuse, accidental changes and leakage. BI security therefore ensures that only the right people can sign in, that each person can do only what their role requires, and that each person sees only the data they are entitled to see.

This experiment applies these ideas to the Global Superstore dataset in Microsoft Power BI. Users and roles are identified, authentication methods are compared, an Access Control Matrix is designed, region-based Row-Level Security is implemented with DAX filters in Power BI Desktop and tested using View As, and a Security Risk Assessment Matrix is prepared.

3. Objectives

1. To identify the organizational users of a BI system and the information each of them requires.
2. To prepare a Role Matrix that links user roles, required reports and access purpose.
3. To study and compare authentication methods: Username and Password, Organizational Account and Multi-Factor Authentication (MFA).
4. To design an Access Control Matrix defining View, Edit, Share and Admin permissions for each role.
5. To plan an RLS strategy and implement region-based RLS roles with DAX filters in Power BI Desktop, and test them using View As.
6. To identify security risks and prepare a Security Risk Assessment Matrix with recommended controls.

4. Software and Tools Required

- **Microsoft Power BI Desktop:** used for loading data, creating matrix tables, defining RLS roles and testing them with View As.
- **Microsoft Power BI Service:** relevant only for tasks performed after publishing, such as assigning users to RLS roles and sharing. No Power BI Service screenshots form part of the evidence for this experiment.
- **Global Superstore Dataset.xlsx:** the source dataset (Orders, People and Returns tables).
- **Microsoft Excel:** the format of the source dataset.

5. Theoretical Background

5.1 Authentication

Authentication is the process of verifying that a user is who they claim to be. In Power BI, users normally sign in with an organizational (Microsoft) account, which is managed by the organization's identity service. Common methods are a username and password, an organizational account, and Multi-Factor Authentication (MFA), where the user proves identity with an additional factor such as a code or an approval on a registered device. MFA is enforced through the organization's identity and tenant settings, not inside Power BI Desktop.

5.2 Authorization

Authorization decides what an authenticated user is allowed to do. Authentication answers "Who are you?", while authorization answers "What may you do?". A user can be authenticated successfully and still be denied an action if they are not authorized for it. In a BI context, authorization determines whether a user may view, edit, share or administer content.

5.3 Role-Based Access Control

Role-Based Access Control (RBAC) assigns permissions to roles rather than to individual users, and users receive permissions by being given a role. This simplifies administration,

keeps access consistent, and supports the principle of least privilege, under which each role gets only the minimum access needed for its work.

5.4 Access Control

Access control is the set of mechanisms that regulate who can view, modify and share resources. In Power BI, access is controlled at several levels, for example workspace roles, sharing of reports and dashboards, and Row-Level Security. An Access Control Matrix is a planning tool that tabulates users or roles against permissions. Creating the matrix documents the intended design but does not by itself assign any permissions in Power BI Service.

5.5 Row-Level Security

Row-Level Security (RLS) restricts the rows of data that a user can see within a semantic model. Roles are created in Power BI Desktop (Modeling → Manage roles), and each role contains DAX filter expressions on tables that return True or False for every row. The roles can be tested in Power BI Desktop using View As. After the report is published, users or groups are assigned to the roles in Power BI Service. In general, RLS restricts users who have Viewer access and does not restrict workspace members with edit rights. When a user belongs to more than one role, the filters of those roles are combined, so the user sees the rows allowed by any of them.

5.6 Data Protection in Business Intelligence

Data protection in BI aims to preserve the confidentiality, integrity and availability of information. It is achieved by layering controls: strong authentication to establish identity, role-based authorization to limit actions, Row-Level Security to limit visible data, controlled sharing to prevent leakage, and periodic review of risks and permissions. No single control is sufficient; the controls complement each other.

6. Implementation and Analysis

Dataset Preparation

The Global Superstore Dataset.xlsx file was loaded into Power BI Desktop (Home → Excel workbook → Load). The Data pane in the screenshots lists the Orders, People and Returns tables, and the Orders table includes fields such as Product ID, Product Name, Profit, Quantity, Region, Row ID, Sales, Segment, Ship Date, Ship Mode, Shipping Cost, State and Sub-Category. The Region column of Orders is used for RLS.

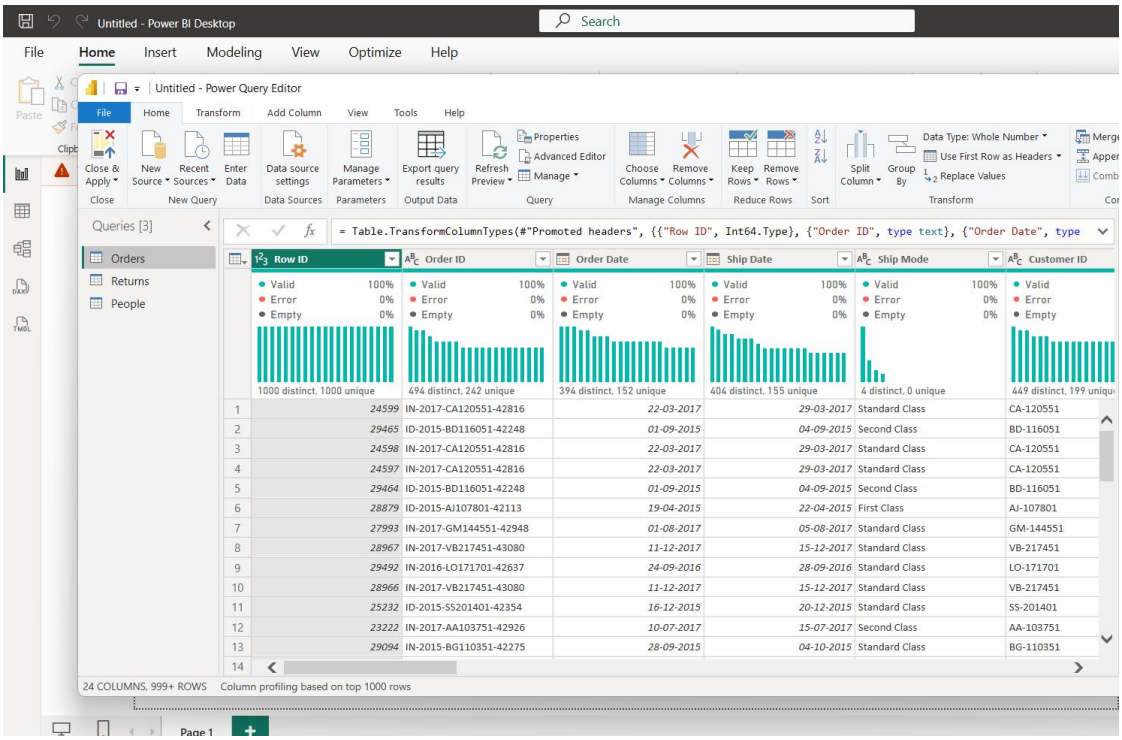


Figure 1: Global Superstore Dataset Loaded in Power BI Desktop

A. User and Role Identification

Four organizational users of the BI system were identified: the CEO, Sales Manager, Finance Manager and Regional Manager. Their needs differ. The CEO monitors overall business performance through the Executive Dashboard. The Sales Manager works with the Sales Report and Customer Report. The Finance Manager analyses revenue and profitability through the Profit Report and Financial Dashboard. The Regional Manager monitors the assigned region through the Regional Sales Report. A fifth role, the BI Administrator, appears in the Access Control Matrix because it manages the BI environment.

The data was entered in Power BI Desktop using Home → Enter data and appears in Power Query Editor as the query Role_Matrix. The profile in Figure 2 shows four distinct, unique User Roles values, with 100% valid values and no errors or empty cells.

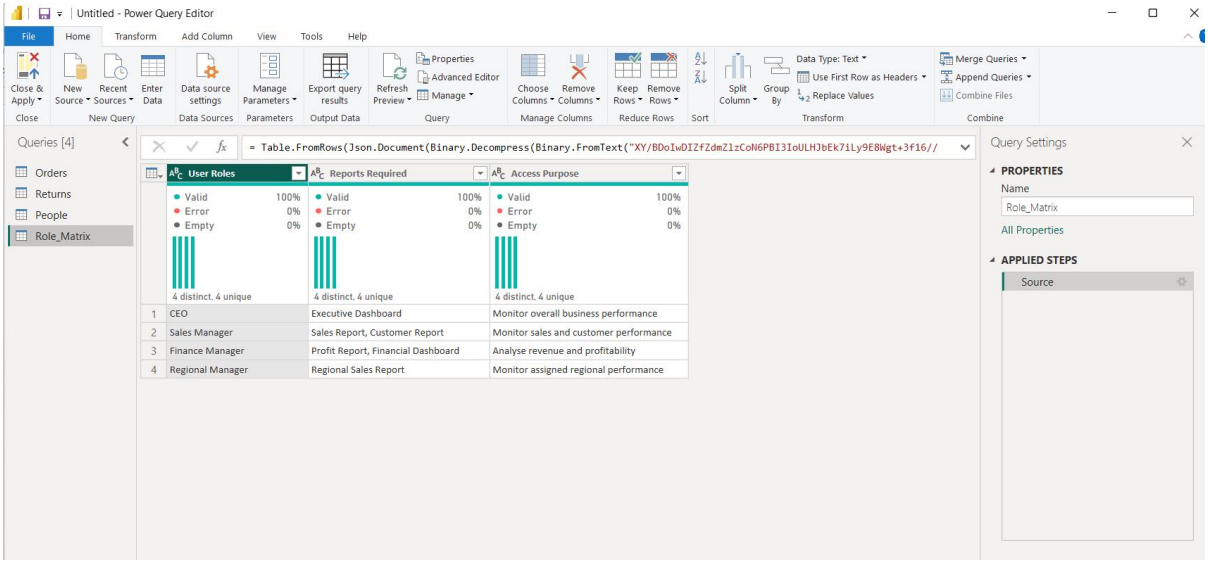


Figure 2: Role_Matrix Table in Power Query Editor

Role Matrix

User Role	Reports Required	Access Purpose
CEO	Executive Dashboard	Monitor overall business performance
Sales Manager	Sales Report, Customer Report	Monitor sales and customer performance
Finance Manager	Profit Report, Financial Dashboard	Analyse revenue and profitability
Regional Manager	Regional Sales Report	Monitor assigned regional performance

The table was loaded into the model and displayed in Report view as a Table visual (Figure 3).

Access Purpose	Reports Required	User Roles
Monitor overall business performance	Executive Dashboard	CEO
Analyse revenue and profitability	Profit Report, Financial Dashboard	Finance Manager
Monitor assigned regional performance	Regional Sales Report	Regional Manager
Monitor sales and customer performance	Sales Report, Customer Report	Sales Manager

Figure 3: Role Matrix

B. Authentication Analysis

Three authentication methods were compared and recorded in the Authentication_Methods table created using Enter data. This is a documented comparison; no authentication setting, including MFA, was enabled or configured in Power BI Desktop.

Advantages	Authentication Method	Description
Centralized identity management	Organizational Account	Users access Power BI using their organization's Microsoft account
Provides an extra layer of security	Multi-Factor Authentication (MFA)	Users verify their identity using an additional authentication method
Simple and easy to use	Username and Password	Users sign in using their login credentials

Figure 4: Authentication Methods Comparison

Authentication Methods Comparison

Authentication Method	Description	Advantages
Username and Password	Users sign in using their login credentials	Simple and easy to use
Organizational Account	Users access Power BI using their organization's Microsoft account	Centralized identity management
Multi-Factor Authentication (MFA)	Users verify their identity using an additional authentication method	Provides an extra layer of security

Username and Password is simple and easy to use but, on its own, gives the least assurance of identity. An Organizational Account centralizes identity management, so access is controlled and reviewed in one place. MFA provides an extra layer of security because a stolen password alone is not enough to sign in.

Advantages of Secure Authentication



Figure 5: Advantages of Secure Authentication

- Prevents unauthorized users from accessing organizational data.
- Protects confidential business information.

• Reduces the risk of account misuse.

• Provides an additional security layer through MFA.

• Helps maintain secure access to reports and dashboards.

Secure authentication is the foundation of the other controls in this experiment: permissions and RLS are meaningful only if the identity of the user is reliably established.

C. Authorization and Access Control

Authorization was designed using an Access Control Matrix (the Access_Control_Matrix table). Four permissions were considered:

• **View:** open and read reports and dashboards.

• **Edit:** modify report content, such as visuals and measures.

• **Share:** give other users access to reports or dashboards.

• **Admin:** manage the BI environment, including user access.

Admin	Edit	Share	User Role	View
No	No	No	CEO	Yes
No	No	No	Regional Manager	Yes
No	Yes	No	Finance Manager	Yes
No	Yes	No	Sales Manager	Yes
Yes	Yes	Yes	BI Administrator	Yes

Figure 6: Access Control Matrix

Access Control Matrix

User Role	View	Edit	Share	Admin
CEO	Yes	No	No	No
Sales Manager	Yes	Yes	No	No
Finance Manager	Yes	Yes	No	No
Regional Manager	Yes	No	No	No
BI Administrator	Yes	Yes	Yes	Yes

All roles have View. The Sales Manager and Finance Manager also have Edit, the CEO and Regional Manager are view-only, and only the BI Administrator has Edit, Share and Admin together.

Justification of Permission Assignments

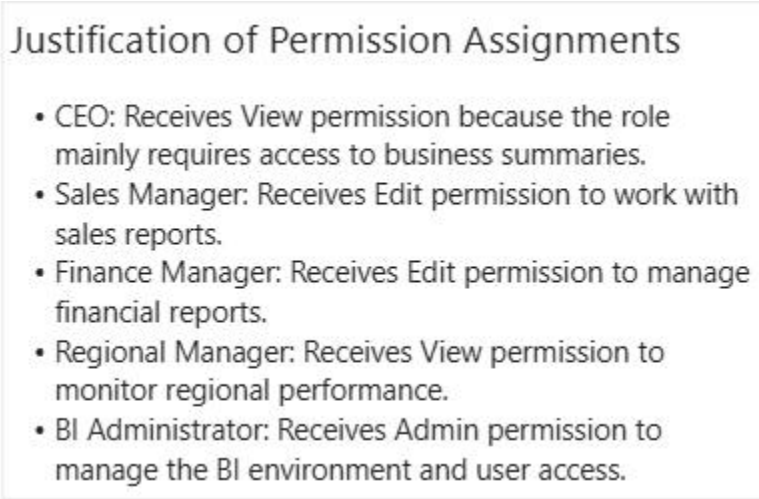


Figure 7: Justification of Permission Assignments

- **CEO:** Receives View permission because the role mainly requires access to business summaries.
- **Sales Manager:** Receives Edit permission to work with sales reports.
- **Finance Manager:** Receives Edit permission to manage financial reports.
- **Regional Manager:** Receives View permission to monitor regional performance.
- **BI Administrator:** Receives Admin permission to manage the BI environment and user access.

The assignments follow RBAC and least privilege: managers who prepare sales or financial reports can edit them, executives and regional managers who only monitor performance can view, and the administrative permissions are limited to one role.

D. Row-Level Security Implementation

RLS Concept and Strategy

Report-level permissions decide whether a user may open a report, but they do not limit which rows the user sees. RLS closes this gap. The strategy was that each regional manager should see only the rows of the assigned region, so the **Region** column of the **Orders** table is filtered. One role is created per region, and the CEO, who monitors the overall business, is designated for all regions in the design matrix.

RLS Design Matrix



Assigned Region	Data Access	User Role
All Regions	All regional data	CEO
Eastern Asia	Eastern Asia only	Eastern Asia Manager
Eastern Europe	Eastern Europe only	Eastern Europe Manager
North Africa	North Africa only	North Africa Manager
South America	South America only	South America Manager

Figure 9: RLS Design Matrix

Assigned Region	Data Access	User Role
All Regions	All regional data	CEO
Eastern Asia	Eastern Asia only	Eastern Asia Manager
Eastern Europe	Eastern Europe only	Eastern Europe Manager
North Africa	North Africa only	North Africa Manager
South America	South America only	South America Manager

The RLS_Design table is a documentation table. The figure above is cropped from the screenshot taken during View As testing. Four of its five rows correspond to the RLS roles implemented below. The "All Regions" row for the CEO is documented in the matrix only; no CEO role appears in the Manage security roles screenshots.

Roles Created in Power BI Desktop

Roles were created using Modeling → Manage roles → New. Each role was given a filter on the Orders table through Switch to DAX editor. The first screenshot shows North_Africa_Manager created with the Orders table selected.

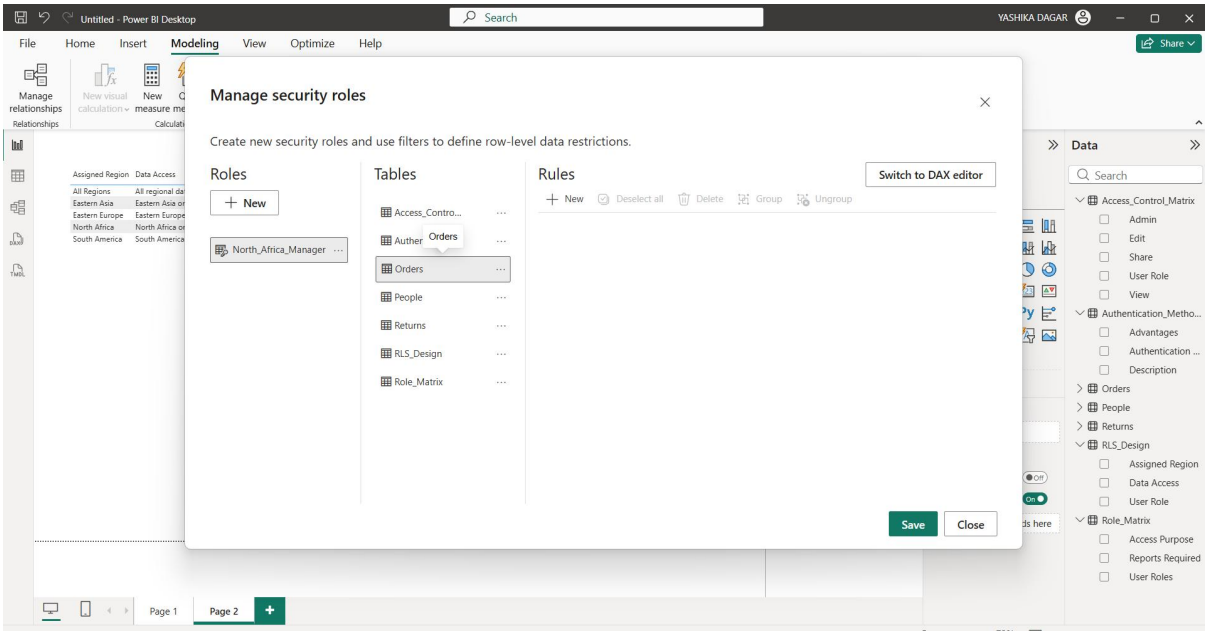


Figure 10: Manage Security Roles: Role Created and Orders Table Selected

Four roles were created, each with a single DAX filter. The expressions below are exactly those visible in the screenshots:

RLS Role	Table	DAX Filter Expression
Eastern_Asia_Manager	Orders	`[Region] = "Eastern Asia"`
Eastern_Europe_Manager	Orders	`[Region] = "Eastern Europe"`
North_Africa_Manager	Orders	`[Region] = "North Africa"`
South_America_Manager	Orders	`[Region] = "South America"`

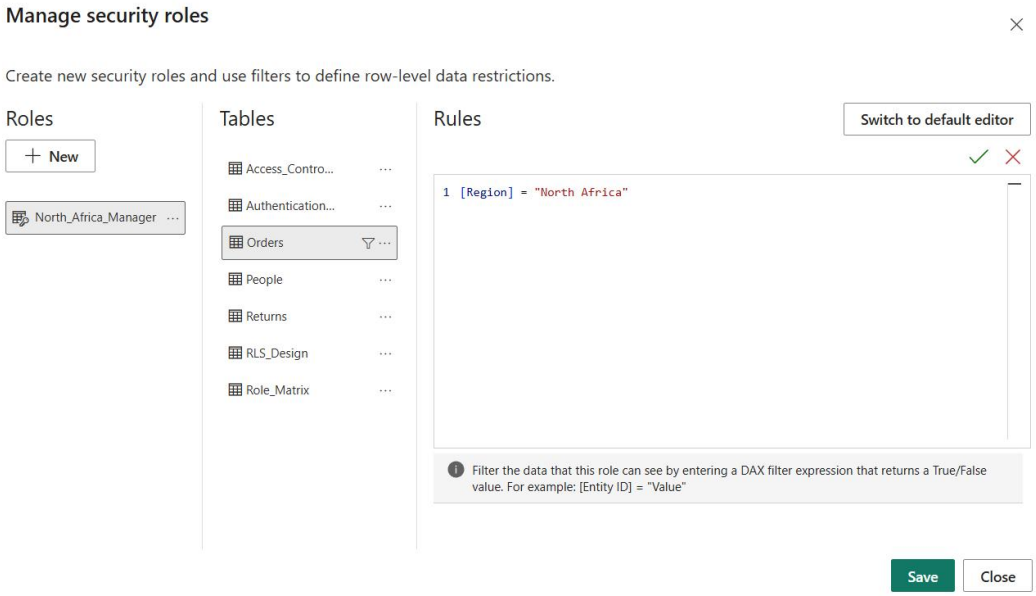


Figure 11: Row-Level Security Filter for North_Africa_Manager

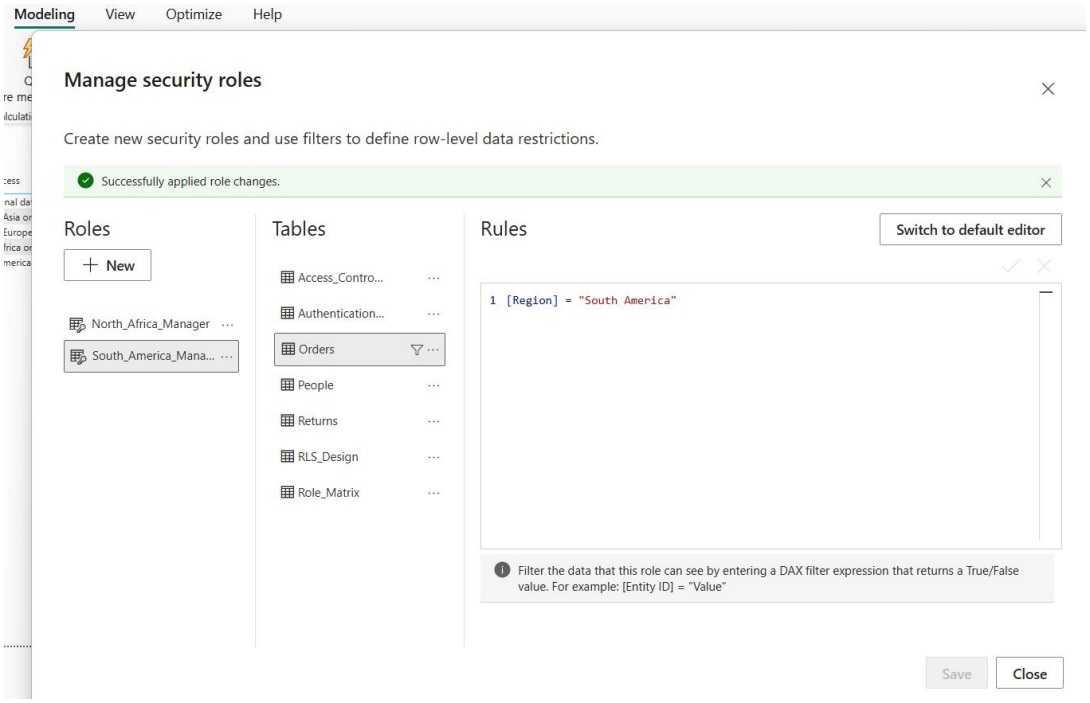


Figure 12: Row-Level Security Filter for South_America_Manager

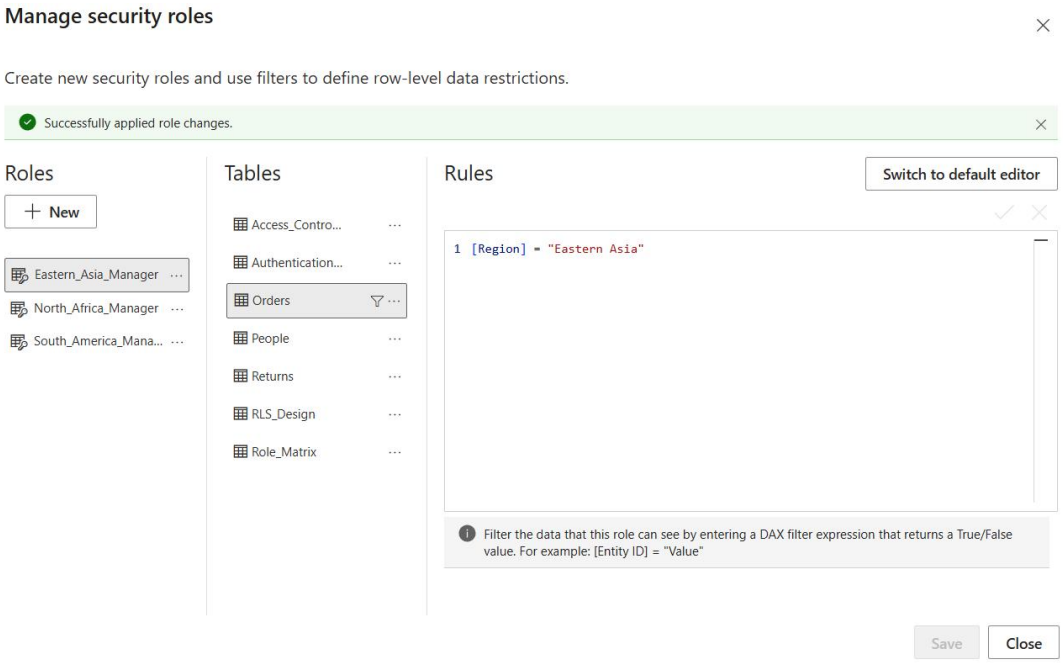


Figure 13: Row-Level Security Filter for Eastern_Asia_Manager

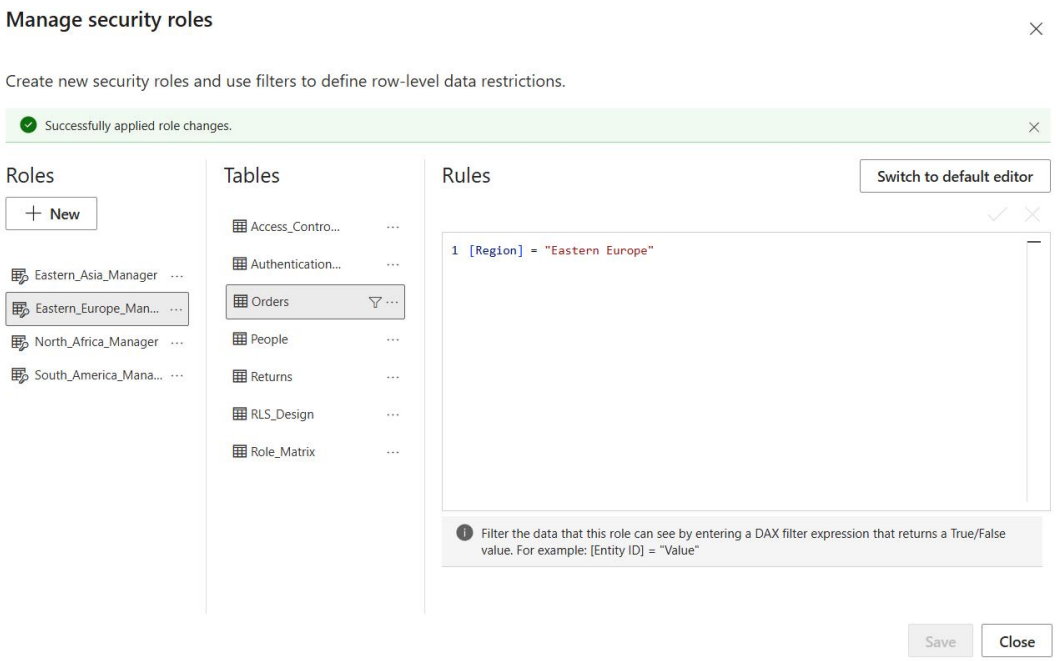


Figure 14: Row-Level Security Filter for Eastern_Europe_Manager

After saving, Power BI Desktop displayed "Successfully applied role changes" and the Orders table showed a filter icon. This is static RLS because each region value is typed directly into the expression.

View As Testing Process

Testing was done using Modeling → View as. In the View as roles dialog (which also offers None and Other user), one or more roles are ticked and confirmed with OK. Report view then shows a "Now viewing as" banner, and Stop viewing returns to normal view. Three role combinations were selected.

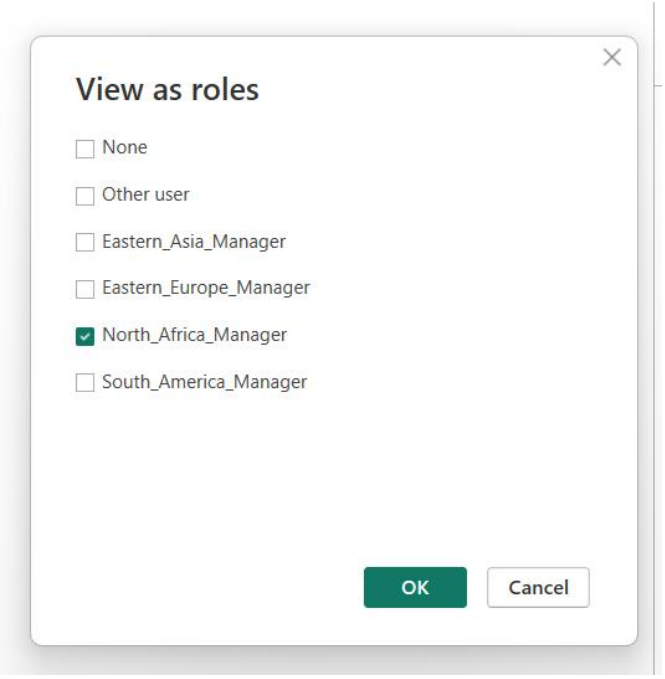


Figure 15: RLS Testing: View as Roles (North_Africa_Manager)

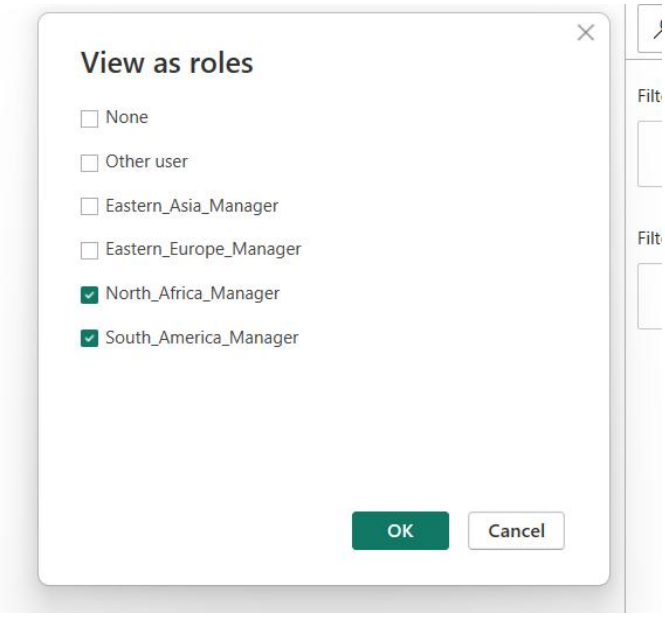


Figure 16: RLS Testing: View as Roles (North_Africa_Manager and South_America_Manager)

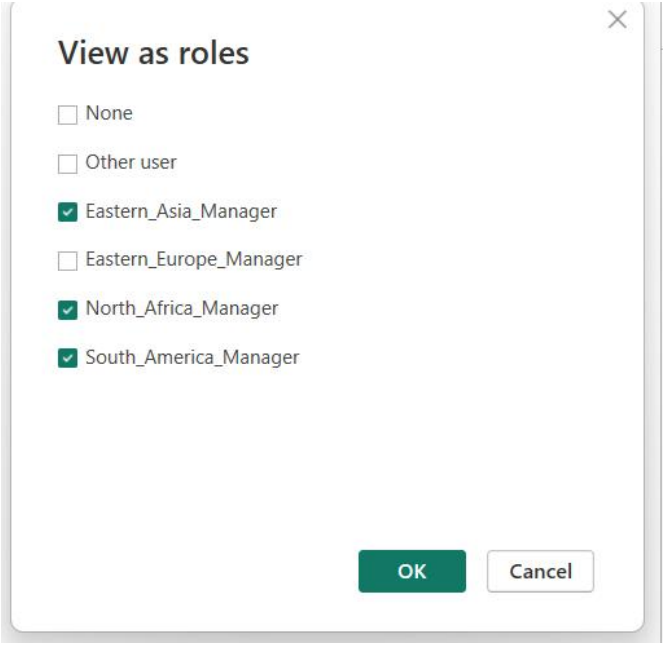


Figure 17: RLS Testing: View as Roles (Eastern_Asia_Manager, North_Africa_Manager and South_America_Manager)

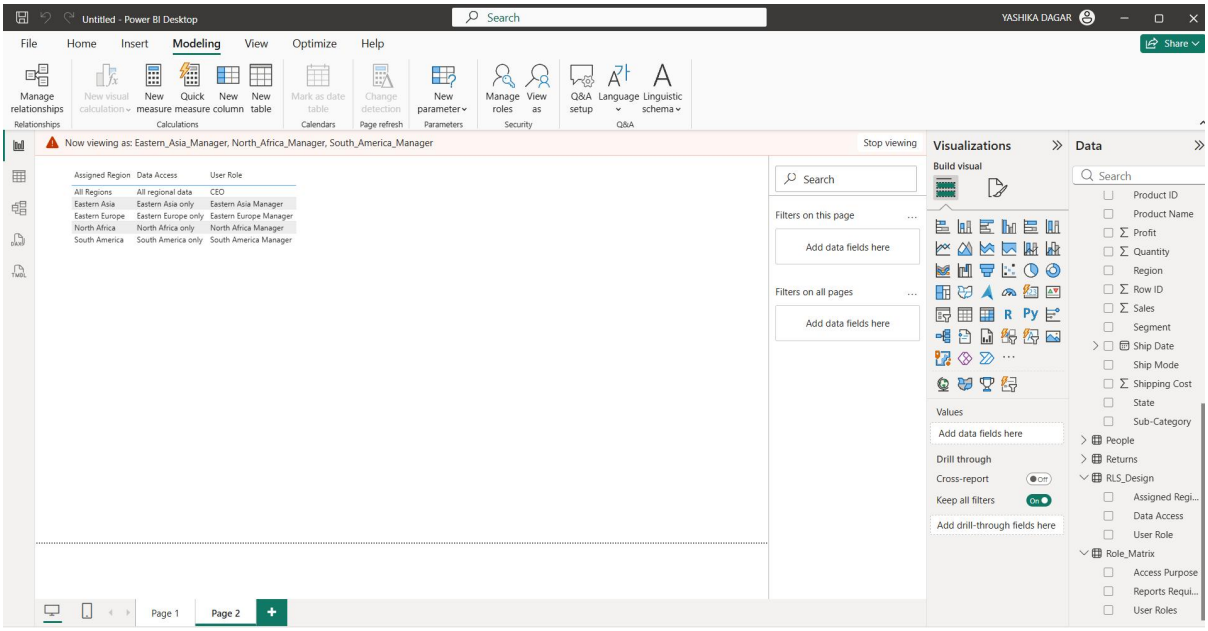


Figure 18: RLS Testing: "Now viewing as" Banner in Report View

Discussion of Actual Filtering Results

The screenshots confirm that the roles could be selected and applied: the banner reads "Now viewing as: Eastern_Asia_Manager, North_Africa_Manager, South_America_Manager". With several roles selected together, the filters are combined, so the user sees rows allowed by any of the selected roles.

The page in the banner screenshot shows only the RLS_Design table, which still lists all five rows. This is expected, because the role filters are defined on the Orders table and not on

RLS_Design. No screenshot of an Orders-based visual under View As was supplied, so the filtered values (for example, sales for a single region) are not reported here and no test result is claimed.

Security Benefits of RLS

- Each manager sees only the data of the assigned region, which protects confidential information.
- One report can serve many users, so separate reports per region are not needed.
- It supports least-privilege access at the row level, in addition to report-level permissions.
- Rules are defined centrally in the data model, which makes them easier to maintain and review.
- It reduces the risk of accidental exposure of other regions' data.

E. Security Risk Assessment

Four risks were identified and recorded in the Security_Risk_Matrix table, created using Enter data (Figure 20) and displayed in Report view (Figure 21).

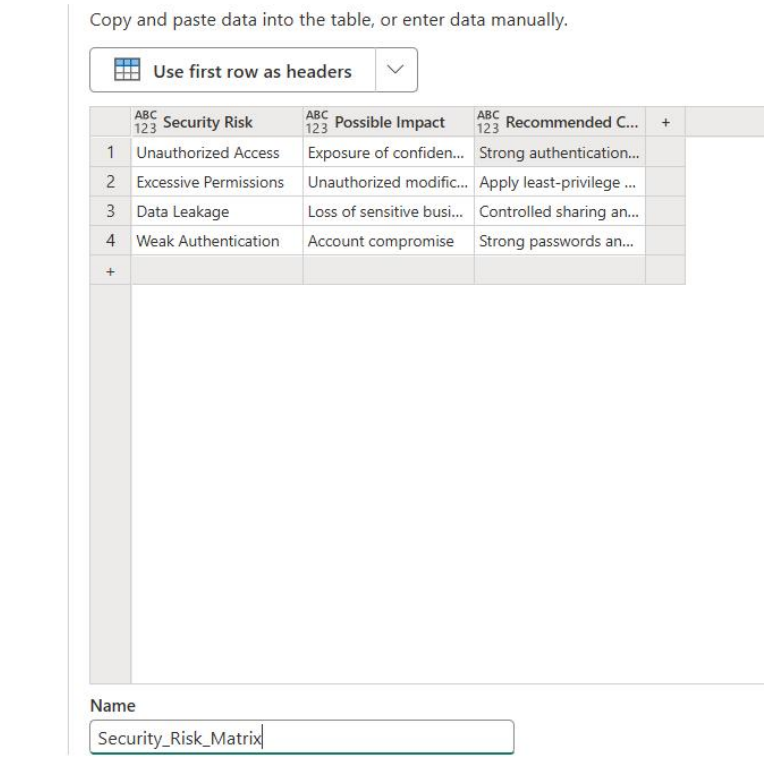
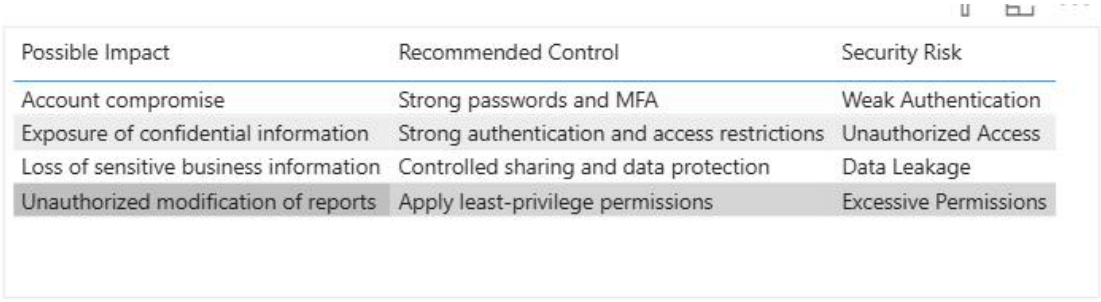


Figure 20: Creating the Security_Risk_Matrix Table Using Enter Data



Possible Impact	Recommended Control	Security Risk
Account compromise	Strong passwords and MFA	Weak Authentication
Exposure of confidential information	Strong authentication and access restrictions	Unauthorized Access
Loss of sensitive business information	Controlled sharing and data protection	Data Leakage
Unauthorized modification of reports	Apply least-privilege permissions	Excessive Permissions

Figure 21: Security Risk Assessment Matrix

Security Risk Assessment Matrix

Security Risk	Possible Impact	Recommended Control
Unauthorized Access	Exposure of confidential information	Strong authentication and access restrictions
Excessive Permissions	Unauthorized modification of reports	Apply least-privilege permissions
Data Leakage	Loss of sensitive business information	Controlled sharing and data protection
Weak Authentication	Account compromise	Strong passwords and MFA

Identified Risks and Controls

- **Unauthorized Access:** a person without a legitimate need reaches reports or data, exposing confidential information. Controls: strong authentication and access restrictions.
- **Excessive Permissions:** a user holds more permissions than their role requires, allowing unauthorized modification of reports. Control: apply least-privilege permissions.
- **Data Leakage:** sensitive information leaves the intended audience, causing loss of sensitive business information. Controls: controlled sharing and data protection.
- **Weak Authentication:** weak credentials allow account compromise. Controls: strong passwords and MFA.

Security Recommendations

The following recommendations are drawn from the Recommended Control column of the matrix and from the RLS design:

- Use strong authentication and access restrictions to prevent unauthorized access.
- Apply least-privilege permissions so that users do not hold more access than their role needs.

• Use controlled sharing and data protection to prevent data leakage.

• Use strong passwords and MFA to reduce the risk of account compromise.

• Apply Row-Level Security so that each manager can see only the assigned region's data.

These are proposed controls. MFA enablement and workspace permission settings were not configured as part of this experiment.

7. Observations and Analysis

1. The Role Matrix shows that each of the four roles needs a different set of reports, which justifies different access levels.
2. The authentication comparison shows a trade-off between simplicity (Username and Password) and assurance (Organizational Account with MFA).
3. In the Access Control Matrix, only the BI Administrator holds all four permissions. The CEO and Regional Manager are view-only, and the Sales Manager and Finance Manager can edit but not share or administer.
4. Four RLS roles were created, each with one DAX filter on the Region column of Orders. Power BI Desktop confirmed each save with "Successfully applied role changes".
5. View As accepted single and multiple role selections, and the banner listed the selected roles. The RLS_Design table was unaffected because the filters apply to Orders only. Filtered Orders output was not captured, so it is not reported.
6. The Security Risk Assessment Matrix links each risk to its impact and control, and the controls correspond to the earlier parts: authentication, permissions and RLS.
7. Limitations: MFA, workspace and sharing permissions, and assignment of users to RLS roles in Power BI Service were not configured or evidenced; the authentication and access control parts are documented analyses.

8. Result

The Business Intelligence security concepts were studied and applied using Microsoft Power BI Desktop with the Global Superstore dataset:

- Four organizational roles (CEO, Sales Manager, Finance Manager and Regional Manager) were identified, and a Role Matrix of reports and access purpose was prepared.
- Username and Password, Organizational Account and MFA were compared, and the advantages of secure authentication were documented. This was a documented analysis; MFA was not configured.
- An Access Control Matrix assigning View, Edit, Share and Admin permissions to five roles (including BI Administrator) was prepared, with justification. It is a design document; no workspace or sharing permissions were configured.
- An RLS Design Matrix was prepared, and four RLS roles (Eastern_Asia_Manager, Eastern_Europe_Manager, North_Africa_Manager and South_America_Manager) were created in Power BI Desktop, each with a DAX filter on [Region] in the Orders table.
- The roles were tested using View As in Power BI Desktop, and the "Now viewing as" banner was displayed. Assignment of users to roles in Power BI Service was not carried out as part of the supplied evidence.
- A Security Risk Assessment Matrix covering four risks, with impacts and controls, was prepared along with security recommendations.

9. Learning Outcomes

After completing this experiment, the student is able to:

1. Identify BI users and define their information requirements using a Role Matrix.
2. Distinguish authentication from authorization and compare common authentication methods, including MFA.
3. Design an Access Control Matrix with View, Edit, Share and Admin permissions based on role requirements and least privilege.

4. Create RLS roles with DAX filter expressions in Power BI Desktop and test them using View As.
5. Explain that RLS roles are created and tested in Power BI Desktop, while users are assigned to roles in Power BI Service.
6. Identify BI security risks and prepare a risk assessment matrix with appropriate controls and recommendations.

10. Conclusion

This experiment showed that BI security works in layers. Authentication establishes who the user is, authorization defines what the user may do, Row-Level Security limits which rows the user can see, and risk assessment ensures that controls match the threats. In Power BI Desktop, four region-based RLS roles were created with DAX filters on the Region column of the Orders table and selected through View As. The Access Control Matrix and the authentication comparison were prepared as documented designs; MFA, workspace permissions and assignment of users to roles in Power BI Service would be carried out separately in a production environment. Together, these controls help protect confidential business information while still giving each user the data they need.